

DRINKING WATER

Mapping safe drinking water use in low- and middle-income countries

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Safe drinking water access is a human right, but data on safely managed drinking water services (SMDWS) is lacking for more than half of the global population. We estimate SMDWS use in 135 low- and middle-income countries (LMICs) at subnational levels with a geospatial modeling approach, combining existing household survey data with available global geospatial datasets. We estimate that only one in three people used SMDWS in LMICs in 2020 and identified fecal contamination as the primary limiting factor affecting almost half of the population of LMICs. Our results are relevant for raising awareness about the challenges and limitations of current global monitoring approaches and demonstrating how globally available geospatial data can be leveraged to fill data gaps and identify priority areas in LMICs.

Access to safe drinking water is recognized as a basic human right and plays a core role in the United Nations' (UN's) 2030 Agenda for Sustainable Development. However, the World Health Organization (WHO) and United Nations Children's Fund Joint Monitoring Programme for Water Supply, Sanitation and Hygiene (JMP) estimate that at least 25% of the global population lacked safe drinking water services in 2020 (1). Addressing the global lack of access to safe drinking water requires that data are available to inform evidence-based planning and resource allocation (2). Nationally representative household surveys provide critical information about populations using "safely managed drinking water services" (SMDWS), which is the global indicator for estimating progress toward the Sustainable Development Goal (SDG) for safe drinking water for all (SDG 6.1). Global use of SMDWS is estimated by the JMP using data on four subcomponents which measure whether households use a primary drinking water source that is (i) improved (3), (ii) accessible on premises (4), (iii) available when needed (5), and (iv) free of fecal and priority chemical contamination (6, 7). However, progress on data collection is slow, global estimates are lacking for approximately half of the global population, and there is limited disaggregation at subnational scales (8).

Safe drinking water is influenced by a range of interacting environmental and socioeconomic

factors. At the landscape scale, water availability can be influenced by local precipitation, evapotranspiration, soil moisture, vegetation, water storage dynamics, and human water use (9). At finer scales, drinking water quality can be influenced by a range of human activities that can be predicted on the basis of population density, livestock ownership, cropland, and forest coverage (10–12). Disentangling the relative importance of these interacting environmental and socioeconomic factors is critical for improving our understanding of SMDWS in different regions. The emergence of geospatial data derived from Earth observation (EO) (including satellite, airborne, and land-based data with global coverage) on human, geographic, and environmental factors provides an opportunity to complement household survey data and generate estimates for progress in sustainable development in data scarce regions (13, 14). However, until now, a spatially explicit evaluation of predictors of SMDWS and its subcomponents at a global scale has not existed.

We explore human and environmental factors driving the spatial variation in use of SMDWS and generate a global map of subnational (15) estimates of SMDWS use across 135 low- and middle-income countries (LMICs). Use of SMDWS and subcomponent estimates in our training set were calculated at subnational district levels using Multiple Indicator Cluster Survey (MICS) data (16, 17) from 64,723 household survey responses, gathered from 27 countries between 2016 and 2020 (Fig. 1 and data S1). Our approach to calculating SMDWS use differs from the current JMP approach in how we integrate the four SMDWS subcomponents at household level rather than at the level of the entire urban and rural population of a country (10). We use a geospatial random forest modeling approach combining SMDWS use estimates from 318 subnational districts (data S1) with 39 geospatial covariates (18–32) (data S2 and

Fig. 1). Further, we determine which of SMDWS subcomponents limit use of SMDWS around the world, using *Escherichia coli* as the measure of drinking water quality due to a lack of globally available data on other drinking water quality indicators at the household level.

Mapping use of SMDWS in LMICs

Overall, we estimate that 33% (interquartile range 32 to 38%) of the total population of 135 LMICs used SMDWS in 2020 (Fig. 2A). Of these, ~61% resided in Asia, 25% in Africa, 11% in the Americas, and 3% in Europe, with the largest number of people concentrated in Southern Asia, sub-Saharan Africa, and Eastern Asia (Figs. 2B and 3A). Our estimate of approximately 4.4 billion people in LMICs lacking SMDWS is more than twice as high as the global JMP estimate of 2 billion people in 2020 (7). Our results fill national data gaps on SMDWS use for 70 countries as of April 2021, representing 65% of the population in LMICs. The majority of LMICs lacking data were in sub-Saharan Africa ($n = 35$) and Latin America and the Caribbean ($n = 15$), but also included countries in Asia ($n = 11$), Europe ($n = 3$), and Oceania ($n = 6$) (data S3). Sub-Saharan Africa included the world's lowest national and subnational rates of SMDWS use (Fig. 3B). In 12 sub-Saharan African countries, <10% of the populations used SMDWS (data S3) and in 89 subnational regions <5% of the populations used SMDWS (data S4).

Our best-performing model used 39 covariates of both human and environmental features derived from EO data and performed well, capturing 53% of the spatial variation in SMDWS based on leave-one-country-out cross-validation [mean absolute error (MAE) = 0.13, coefficient of determination (R^2) = 0.53] (fig. S1)]. When compared to estimates reported by the JMP at the national level, which included data from administrative and other data sources, our predictions were most aligned in African countries (MAE = 0.13) but were systematically lower than JMP estimates in the Americas (MAE = 0.36), Asia (MAE = 0.23), Europe (MAE = 0.36), and Oceania (MAE = 0.34) (fig. S2). This may be influenced by different methodologies used to calculate the proportion of populations using SMDWS. The methodology used by the JMP takes a minimum of at least three subcomponent estimates (which must include microbial drinking water quality) at the level of the entire urban and rural population. By contrast, we determine SMDWS use at the household level and then calculate the proportion of populations using SMDWS and only include observations with information on all four subcomponents. This leads to systematically lower estimates than the JMP approach. Differences may also be driven by the unequal distribution of geographic regions



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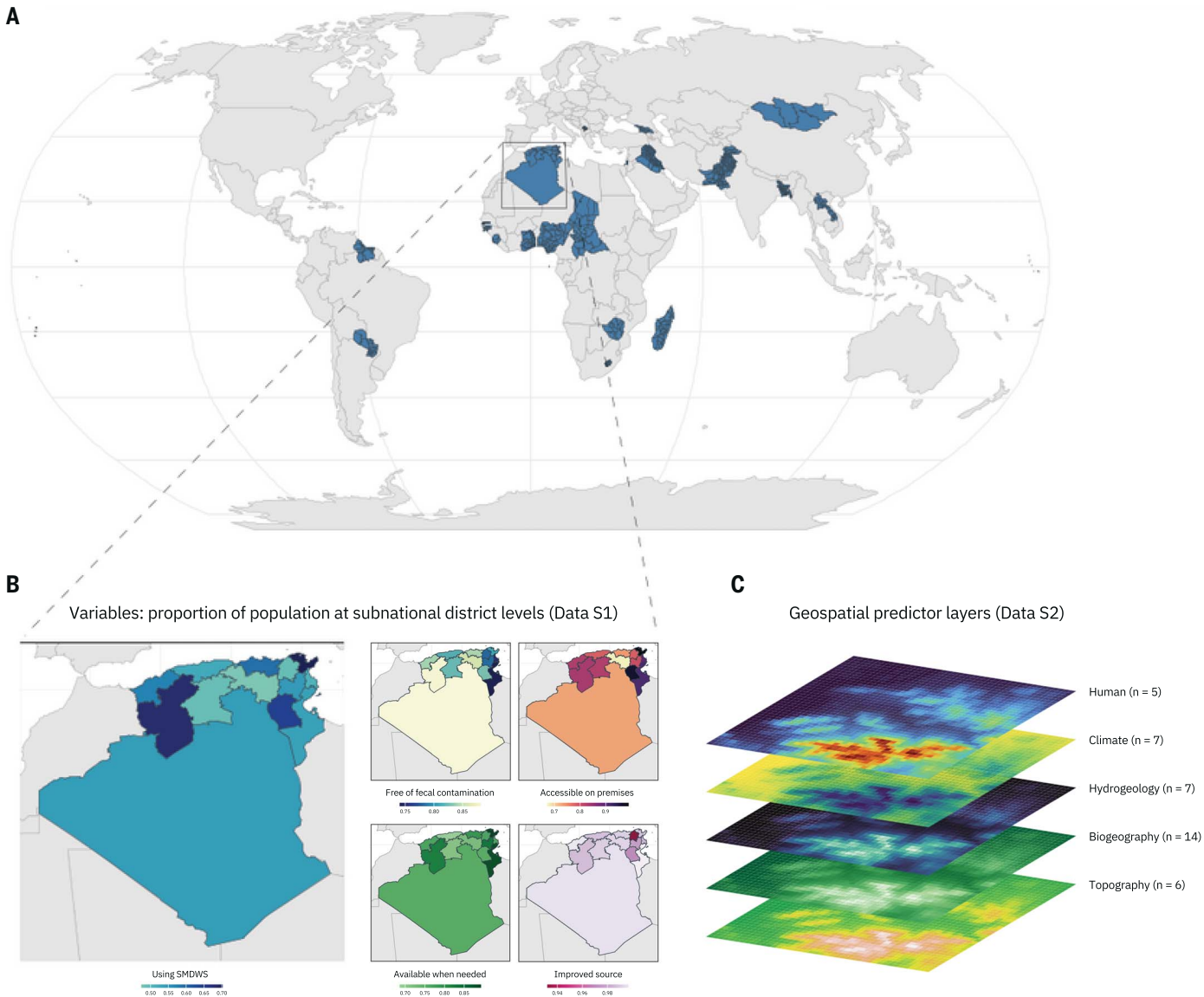


Fig. 1. Overview of data sources and data merging. (A) SMDWS use and subcomponent estimates in 27 low- and middle-income countries were calculated at subnational district levels using 64,723 household survey responses from MICS data. (B) The outcome variable for the main model is the proportion of a district population using SMDWS. Subcomponent outcome variables are the proportion of a district population using a drinking water source which is (i) free

of fecal contamination, (ii) accessible on premises, (iii) available when needed and (iv) improved. (C) Visual representation of geospatial predictors (67), which were merged with the outcome variables at subnational district levels and included human ($n = 5$), climate ($n = 7$), hydrogeologic ($n = 7$), biogeographic ($n = 14$), and topographic ($n = 6$) indicators derived from globally available geospatial datasets (data S2).

and income classes represented in our training and prediction data (fig. S3). Similar differences between JMP estimates by UN region were, however, also found when compared to average predictions from a set of 100 random forest models built on geographically stratified bootstrapped samples of the original training data, which reflected the geographical distribution of LMICs across UN subregions (table S1). We additionally tested the validity of our predictions on a separate test dataset using MICS survey data ($n = 46$ regions) from seven LMICs collected between May 2019 and January 2022, calculating SMDWS at the household level

with all four subcomponents. Predictions from our best-performing model captured 28% of the spatial variation (MAE = 0.14) and predictions from the geographically stratified ensemble model captured 34% of the spatial variation in SMDWS use (MAE = 0.13) (fig. S4). These results point toward potential improvements in prediction accuracy when the geographical distribution of training and prediction data points are similar and ensemble models are used which regularize predictions of several individual models.

Our analysis offers insights into the limiting factors driving lack of SMDWS use in different

regions around the world (Fig. 2, C and D). Four separate distributed random forest models were trained to predict the proportion of subnational populations using a primary drinking water source which was (i) improved, (ii) accessible on premises, (iii) available when needed, and (iv) without fecal contamination as indicated by the absence of *E. coli* in a 100-ml sample (data S5 and Fig. 2D). Models were trained using the same 39 covariates and household surveys as the SMDWS model and were best at predicting drinking water accessibility on premises ($R^2 = 0.48$, MAE = 0.18) (fig. S5) and fecal contamination ($R^2 = 0.44$,

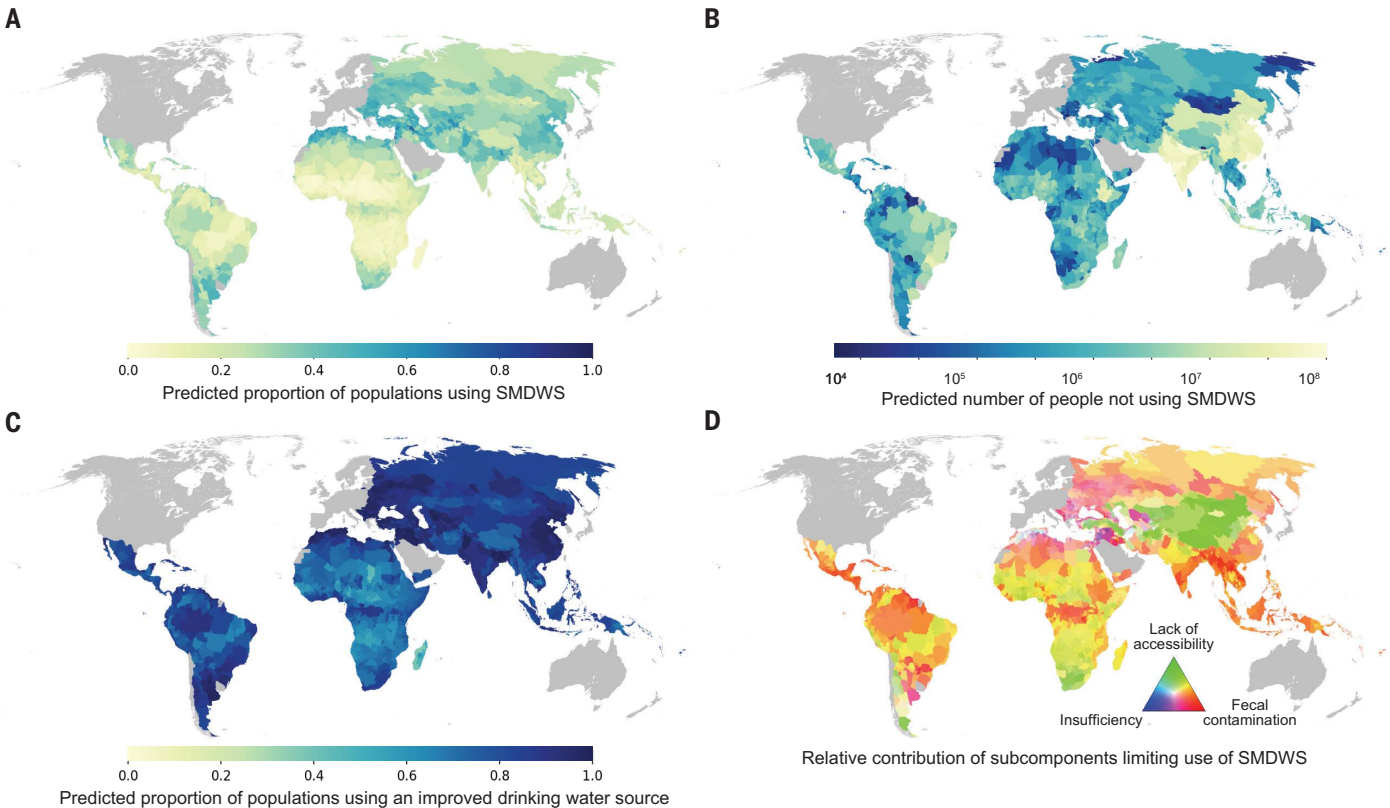


Fig. 2. Mapped SMDWS use. This is shown in terms of population percentage (A) and number of users (B); percentage of people using an improved drinking water source (C) and a map showing where fecal contamination (red), lack of accessibility (green), and insufficiency (blue) of drinking water from a primary drinking water source are limiting use of SMDWS (D) at global administrative area level 1 in LMICS. A combination of colors means that use of SMDWS is limited by more than one of the respective subcomponents (D). Results are depicted from the best-performing models based on leave-one-country-out cross-validation metrics.

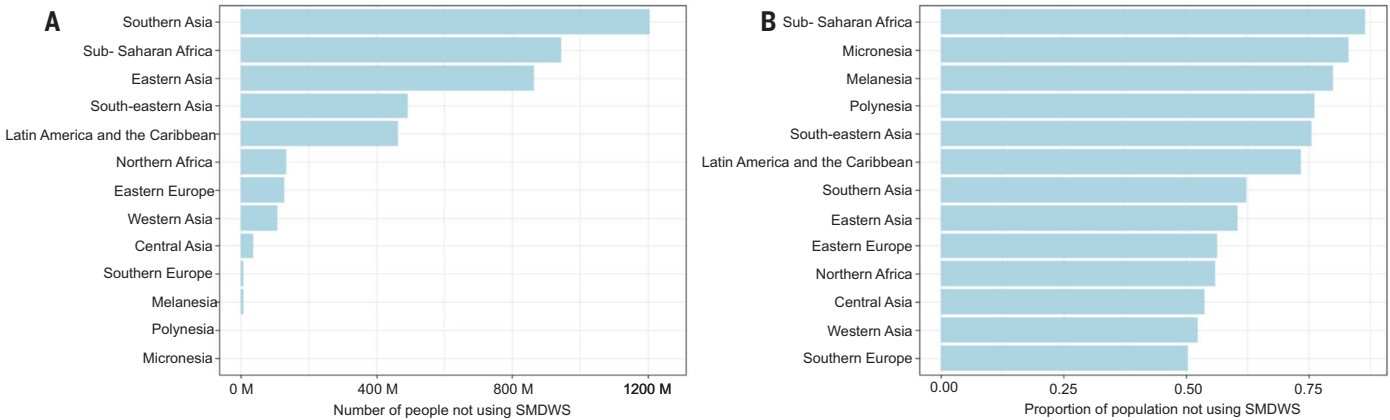


Fig. 3. Distribution of SMDWS use across UN subregions (68). Number of people (in millions) predicted not to use SMDWS (A) and the proportion of populations predicted not to use SMDWS (B).

MAE = 0.16) (fig. S6) followed by an improved drinking water source ($R^2 = 0.25$, MAE = 0.11) (fig. S7). Model performance was worse when predicting populations with drinking water availability ($R^2 = 0.01$, MAE = 0.09) (fig. S8) based on leave-one-country-out cross-validation R^2 . We attribute the low R^2 for the water availability model to the limited range and variability of water availability values (range: 30 to 100%, interquartile range: 80 to 92%) in

the training dataset. However, it could also reflect limited associations between predictors and a household member's subjective assessment of experiencing water insufficiency which is influenced by their practices of water storage or use of multiple water sources when in need (33).

We estimate that 88% of all people living in LMICS use an improved drinking water source, which is defined by the JMP as having “the

potential to deliver safe water by nature of its design and construction”; piped water is one such example (6) (Fig. 2C). Our results emphasize that access to an improved drinking water source does not always provide safe drinking water as almost half of the LMIC populations (48%) were estimated to be exposed to fecal contamination in their primary drinking water source. Our predictions show that more than half of the populations of Oceania, sub-Saharan

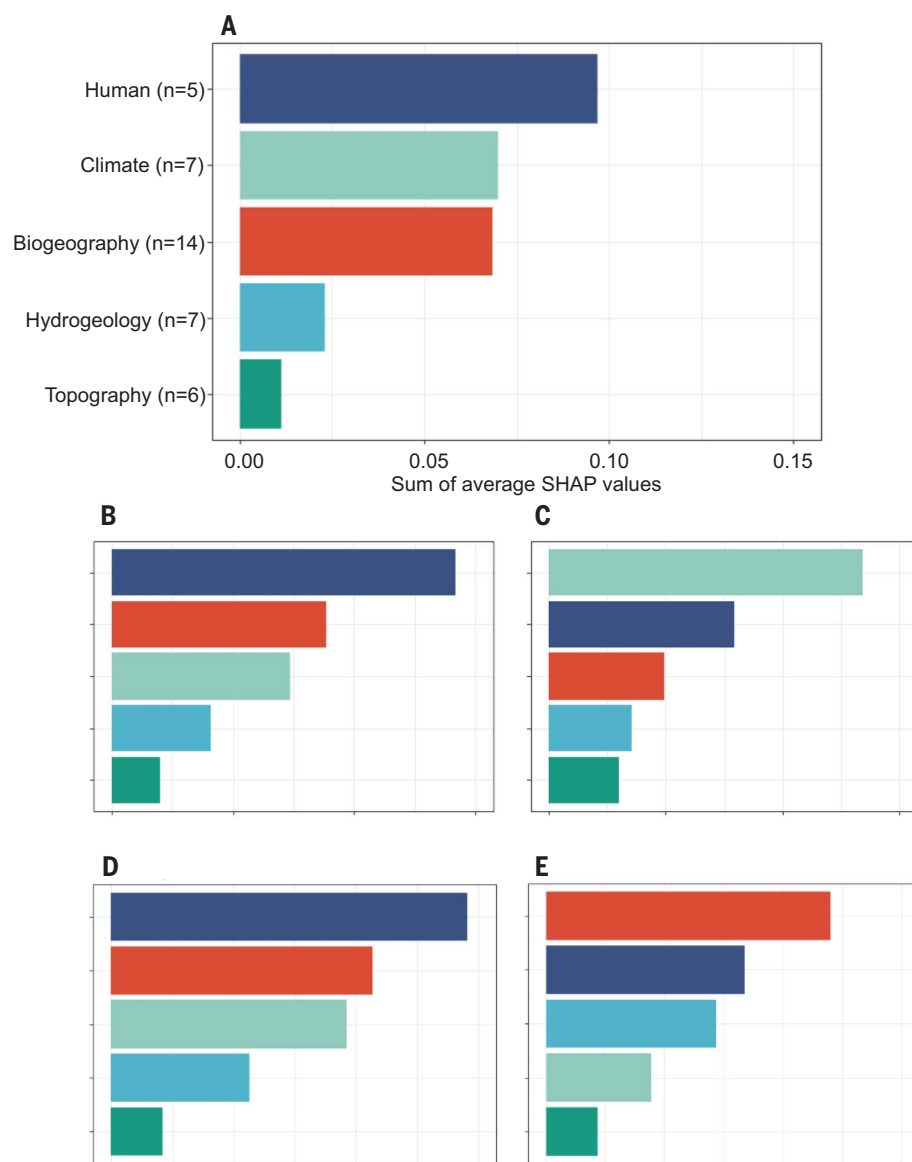


Fig. 4. Grouped and summarized absolute SHAP values for predicting SMDWS use. (A) Improved drinking water source use; (B) absence of fecal contamination of primary drinking water source; (C) accessibility of drinking water source on premises; and (D) drinking water availability when needed (E).

Africa, southeastern Asia, and Latin America and the Caribbean may be exposed to contaminated drinking water (fig. S9). Lack of accessibility of drinking water on premises was the second most common subcomponent limiting safe drinking water coverage, with an estimated 36% of the overall LMIC population affected. Major investments are needed to bring drinking water supplies closer to households, especially in sub-Saharan Africa where we estimate that over 650 million people lack drinking water services on premises. An increase in access to drinking water within the household, yard, or plot is key for reducing poverty and improving gender equality and health, as it reduces the time spent by women collecting water from distant sources (34–36).

Both human and environmental factors were important for predicting use of SMDWS. Shapley additive explanation (SHAP) values provided quantitative insights into the relative importance of predictors on estimates of SMDWS as well as its subcomponents. Human features had the highest overall SHAP values, capturing more of the variation in SMDWS use than biogeographic, climatic, hydrogeologic, or topographic features (Fig. 4A and data S6). Low urban built-up land cover and low population density were dominant human drivers for lower estimates of SMDWS (Fig. 5), aligning with observed inequalities in delivery of safe drinking water services in rural versus urban areas (1, 37, 38). Prompted to investigate the impact of governance indicators on SMDWS

estimates, we added cross-country comparable worldwide governance indicators on regulatory quality and governance effectiveness to the best-performing geospatial model (39) (data S1). Model performance did not improve with the additional governance indicators based on leave-one-country-out cross validation metrics ($R^2 = 0.48$, MAE = 0.13). However, SHAP contributions calculated on the training data showed that higher indicator values for governance effectiveness and regulatory quality were associated with higher estimates of SMDWS (fig. S10). Separate models trained on subsets of the 39 final geospatial predictors showed that environmental factors alone explained more of the spatial variation ($R^2 = 0.45$, MAE = 0.14) in use of SMDWS than human factors ($R^2 = 0.37$, MAE = 0.14), as determined by leave-one-country-out cross-validation metrics. Annual daily average air temperature, along with amounts and seasonal variation in precipitation, showed a clear directional influence on SMDWS estimates, with higher temperature and precipitation values contributing to lower estimates of SMDWS (Fig. 5). Although the annual amount of precipitation was most important for predicting *E. coli* contamination and availability of drinking water, the coefficient of variation of monthly precipitation, which describes seasonality, was important for all subcomponent models (figs. S11 to S14). Despite the associative nature of the modeling approach, other studies have shown that precipitation patterns may affect the SMDWS subcomponents in several ways. While increased precipitation has predominantly been associated with increased *E. coli* contamination of drinking water sources in LMICs (40, 41), reduced rainfall in dry seasons has been associated with lack of water availability, increased distances traveled for water collection, and use of alternative unimproved drinking water sources (42–44). Notably, findings in this study may be affected by the distribution of data across climate zones and seasons in the training set. Of the 64,723 household surveys used in our training data, 38,954 were classified as located in temperate and tropical climates (fig. S15); of these, 33% were collected during or directly after a wet period lasting at least 15 days (data S1 and fig. S16). Further research is needed to evaluate how global estimates on SMDWS and its subcomponents are affected by seasonality and whether increases in annual temperatures and rainfall may have an impact on future SMDWS access (45, 46).

The order of importance of predictor groups differed between the individual SMDWS subcomponent models, with human factors playing the most important role for predicting the type of drinking water source used (Fig. 4B and fig. S11) as well as its accessibility (Fig. 4D and fig. S12). Climate variables, on the other hand, made the largest contribution to the prediction of fecal contamination (Fig. 4C and fig. S13), with higher

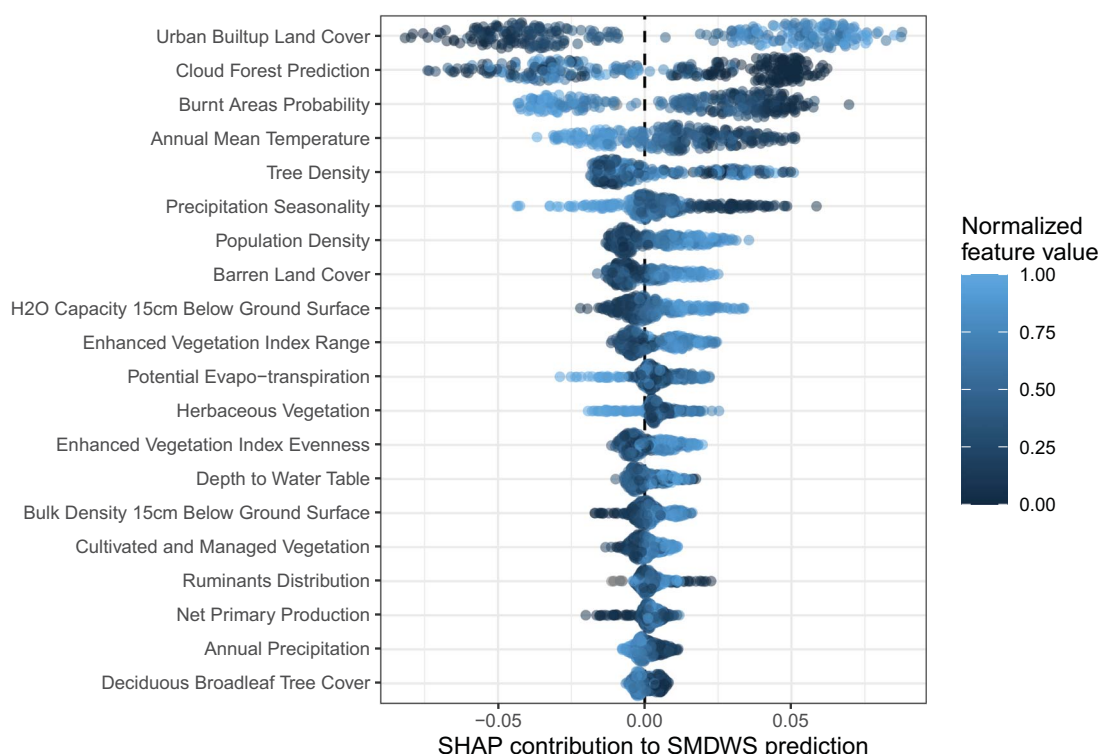


Fig. 5. SHAP contributions of the top 20 out of 39 most important features for predicting SMDWS use for 318 subnational districts (training dataset). Feature importance is depicted on the y-axis with the most important features listed first and SHAP contributions to SMDWS

predictions on the x-axis. Each jitter point represents a subnational district with positive (negative) feature contributions if the point is located above (below) 0 on the x-axis. Light (dark) blue points represent higher (lower) feature values.

annual mean temperatures contributing to higher estimations of *E. coli* contamination reflecting the sensitivity of microbial contamination of drinking water sources to environmental conditions. Biogeographic indicators such as enhanced vegetation index were major contributors for predicting availability of drinking water when needed (Fig. 4E and fig. S14) and point toward associations between water availability for plants and humans. Hydrogeological indicators, primarily depth to bedrock, also played an important role for predicting water availability. These findings underline the interconnected nature of a location's environmental conditions and human development in provision of safe drinking water in LMICs and highlight opportunities for EO-derived data sources to be leveraged.

We excluded high-income countries because of their lack of representation in the training dataset. However, it is likely that some populations within high-income countries also have inadequate access to SMDWS (47, 48). Furthermore, we note that 23% of subnational regions fell outside of the area of applicability (AOA) of our SMDWS model, which refers to areas in which the data used to predict SMDWS estimates were outside of the range of data included in our training set (fig. S17 and data S7). The highest uncertainties estimated

by the interquartile range and coefficient of variation of 100 distributed random forest models built on geographically stratified bootstrapped samples of the training set lie in northern Africa, northern parts of sub-Saharan Africa, and western Asia (fig. S18 and data S4). SMDWS estimates of areas which lie outside the AOA or those with a large range of prediction results are modeled but should not be used as an input for other models to avoid the propagation of modeling errors (49).

Discussion

We have used a geospatial modeling approach combining EO-derived data and household survey data to evaluate the use of SMDWS across 135 LMICs. Our results highlight that SMDWS use in LMICs are primarily limited by fecal contamination, indicated by *E. coli* contamination in the primary drinking water source. *E. coli* quantification is a widely available, technically and economically feasible method for indicating fecal contamination. Nevertheless, monitoring fecal contamination is the most resource-intensive subcomponent. As a result, national data are lacking for countries representing approximately half of the global population (8). Detection of fecal contamination in drinking water is concerning as ingestion of fecal pathogens is a major driver of diarrheal

disease, one of the leading causes of under-five child mortality globally (50, 51). Information derived from cross-sectional *E. coli* data are, however, limited in capturing public health risks related to fecal contamination (52). Multiple sources of drinking water contamination as well as variations in persistence and transport characteristics of bacteria, viruses, and parasites in the environment make risk of pathogen exposure challenging to measure using a single indicator (53). As microbial drinking water quality can vary rapidly, single *E. coli* measurements may be poor indicators of public health risk: Water safety planning and sanitary inspection provide a complementary risk-based approach for preventative management of the water supply system (54, 55). Additionally, the availability of faster, more sensitive, and specific molecular testing methods may provide opportunities to improve drinking water safety monitoring (56). Notably, drinking water contamination and associated risks are only one exposure pathway of pathogen transmission and effective drinking water monitoring and management does not preclude risks of transmission through other routes including soil, food, hands, and flies (57).

Increased availability of higher resolution data on SMDWS for LMICs may improve estimates and allow climate influences to be

captured. The spatial resolution of the majority of drinking water quality data currently made available by national agencies is restricted to Global Administrative Areas (GADM) levels one or two. Georeferenced survey data could support the identification of inequalities in SMDWS use, which may be at risk of increasing with rising urbanization and associated income inequalities in urban areas (38, 58, 59). Multiple data points over time as well as survey questions related to seasonal changes in drinking water use practices could also support the assessment of temporal and seasonal trends in SMDWS use, which current cross-sectional survey data are limited in capturing.

Additionally, we note that we did not include data on priority chemical contamination in our models despite concentrations above 10 µg/L for arsenic and 1.5-mg/L for fluoride being defined as unacceptable as part of the global SMDWS indicator. Priority chemical contamination is only evaluated for a subset of high-risk LMICs and data was insufficient for estimating concentrations in household drinking water across LMICs. Notably, among the countries that included arsenic testing in our training data (Bangladesh and Nepal), *E. coli* contamination was detected significantly more often than arsenic concentrations exceeding the value of 10 µg/L (10) and was more often the limiting factor to SMDWS. Nevertheless, it should be noted that ingestion of arsenic and fluoride can lead to cancer and other health disorders, and global groundwater risk maps should additionally be used to identify regions where household drinking water testing and treatment are needed (60, 61).

Our maps draw attention to a high number of areas potentially lacking SMDWS in LMICs, which may inform the mobilization and effective allocation of financial resources and human capacity. By filling crucial data gaps, our results point toward a substantial underestimation of the number of people whose basic human rights to safe drinking water are not being met and provide information on which subcomponents may be limiting use of SMDWS regionally.

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of its authorities, or concerning the delimitation of its frontiers or boundaries. **Data and materials availability:** Data used in this study were extracted from the following sources: Estimates of SMDWS were calculated using Multiple Indicator Cluster Survey data and are provided by UNICEF through the MICS website (16). Data sources of geospatial predictor variables are documented in Data S2. Country names and codes were extracted from the World Population Prospects Demographic Indicator file for 2022 (62) and information on income groups and Worldwide Governance Indicators downloaded from the World Bank's respective websites (63, 64). National estimates of SMDWS use reported by the JMP for

2020 were downloaded from the JMP website (65). All the code needed to replicate our work as well as the preprocessed geospatial data is publicly available on Zenodo and Github (66). **License information:** Copyright © 2024 the authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original US government works. <https://www.science.org/about/science-licenses-journal-article-reuse>. This research was funded in whole or in part by Swiss National Science Foundation (grant 192763), a cOAlition S organization. The author will make the Author Accepted Manuscript (AAM) version available under a CC BY public copyright license.

SUPPLEMENTARY MATERIALS

[science.org/doi/10.1126/science.adh9578](https://doi.org/10.1126/science.adh9578)
Materials and Methods
Supplementary Text
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